

Lesson 10: The Research Frontier: Regimes, Reinforcement Learning and AI Agents

Analyst onboarding · Lesson 10 of 10 · L/S Gamma Desk, QUANTT

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Goal: understand the advanced research the desk is exploring, what each idea could improve, and how a new idea earns its way into live trading.

You need: Lessons 5–9.

1 Where the live strategy could improve

Our live strategy is a simple **rule**: one forecaster, a fixed band, one structure per side, a daily hedge. Each piece is a place where smarter methods might help:

Piece	Today	Research direction
Forecast	GARCH	ML forecasts, forecast combination with RL
Market state	Not used	Regime models (HMM, jump models)
Which trade	Long call / short put	Choose among 10 structures by regime, RL or LLM
Hedging	Delta hedge, daily	Deep hedging with RL
Sizing	1 contract	Size by regime confidence or risk budget

2 Idea 1: market regimes

A **regime** is a market “mood” that lasts weeks or months: calm, normal or stressed. The best trade differs by regime. Selling volatility works best in calm regimes and worst in crises.

- **Hidden Markov models (HMMs)** assume the market switches between a few hidden states, each with its own volatility. The model outputs **probabilities**, e.g. 80% calm and 20% stressed, rather than a hard label. The PRISM paper sizes positions down when the regime is unclear.
- **Statistical jump models** are clustering with a penalty for switching regimes too often, so regimes stay stable. With no penalty they reduce to **k-means** clustering.
- **Why not plain k-means?** It groups similar-looking days but never sees which trades made money, and it ignores time order, so it flips too often. (See our clustering report.)

3 Idea 2: reinforcement learning (RL)

Reinforcement learning trains an **agent** to make decisions by trial and error in a simulated environment:

1. **State:** what the agent sees, e.g. SPY price, position Greeks, IV, time to expiry.
2. **Action:** what it does, e.g. how many shares to hold.
3. **Reward:** the score after each step, e.g. P&L minus a risk penalty.
4. **Policy:** the rule it learns, a mapping from state to action, usually a neural network.

The agent plays millions of simulated days and gradually learns actions that maximize total reward.

4 Idea 3: deep hedging

Deep hedging (Buehler et al.) replaces “set delta to zero” with an RL policy that learns **how many shares to hold** given costs and risk preferences.

- **What it learns:** to skip hedges that cost more than they save, to stay slightly under- or over-hedged, and to use IV and the VRP in its decisions.
- **For scalping:** a long-gamma hedger could be trained to maximize **scalping P&L after costs**, not just to minimize risk.
- **The danger:** if the reward rewards profit too much, the agent stops hedging and starts **taking directional bets**. Strong penalties on bad outcomes keep it honest.
- **Our plan:** a deep hedger is just another file in `core/Algos/Hedging/` with a `target_shares()` function, compared against plain delta hedging.

5 Idea 4: RL for trading decisions and forecasts

- **OPHR** (a multi-agent RL paper) uses one agent to choose long, short or neutral volatility, and a second agent to choose which hedger to use. That is the same split our code already has.
- **RL forecast combination:** rather than picking GARCH *or* HAR, an RL agent learns how to **weight** several forecasts depending on market conditions.

6 Idea 5: LLM agents as decision makers

Large language models such as Claude can read the day’s features and choose an algo from a fixed menu. The *Agents Are Not Algorithms* paper (2026) found:

- Agents did best when given **pre-computed numbers** and **coded execution rules**, deciding only the judgement calls.
- Agents were **competitive with good rules, not clearly better**.

For us: an LLM may only choose from our menu of structures. It never places orders or hedges directly. We would run it in **shadow mode** first: it logs what it would have done next to the live rule, without trading.

7 How a new idea earns its way into live trading

1. **Write it up:** research the question (our `/strategy-research` workflow produces a sourced PDF in `research/strategy_research/`).
2. **Prototype** in the research sandbox: `research/Deep_Hedging/`, `research/RV_Forecasting/` or `research/Agentic_Routing/`.
3. **Backtest it honestly** (Lesson 9): walk-forward, costs, every trial logged.
4. **Beat the baseline.** It must outperform the simple rule after costs, not just look clever.
5. **Shadow mode:** log its decisions live next to the current rule for several weeks.
6. **Promote it:** plug it into the right registry (`FORECASTERS`, `SIGNALS` or `Hedging/`) and change one line in `configs/paper.toml`.

⇒ Complexity must **pay for itself**. Several of the strongest papers we’ve read found that simple models (linear regressions, threshold rules) capture most of the gains of deep networks.

8 Reading list (in docs/papers/)

- `Deep-Hedging.pdf`: the original deep hedging framework.
- `OPHR-Mastering-Volatility-Trading-with-Multi-Agent-Deep-Reinforcement-Learning.pdf`: RL for long/short vol and hedger choice.
- `PRISM-A-Probabilistic-Regime-Integrated-Scalping-Model-for-Derivatives-Arbitrage.pdf`: regimes and gamma scalping (a whitepaper, simulated results).
- `Agents-Are-not-Algorithms.pdf`: LLM agents in trading.
- `Downside-Risk-Reduction-Using-Regime-Switching-Signals-A-Statistical-Jump-Model-Approach.pdf`: jump-model regimes.
- `Hedging-with-Linear-Regressions-and-Neural-Networks.pdf`: why simple models are a tough baseline.

9 Words to know

- **Regime**: a lasting market state. **HMM**: model of hidden switching states.
- **RL agent / state / action / reward / policy**: the learner, what it sees, what it does, its score, and its learned rule.
- **Deep hedging**: an RL-learned hedging policy.
- **Shadow mode**: running a model live without letting it trade.

10 Check yourself

1. Why might a deep hedger trained only to maximize profit be dangerous?
2. An LLM router beats our rule in a backtest. What are the next two steps before it trades?
3. Why do we keep the rule baseline even after adding ML?

Answers: (1) It can drift into unhedged directional bets. (2) Walk-forward with costs and logged trials, then shadow mode. (3) Every new model must beat it, and it's the fallback if the model fails.